

top-notch even in its smaller, rugged format.

The COM-HPC 1.2 specification, ratified in October 2023, introduced the COM-HPC Mini, thus expanding the standard to accommodate more compact formats. This addition not only escalates the scalability of the COM-HPC standard but also enhances its application range across various high-performance computing environments.

Companies like Kontron have taken immediate steps to innovate within this new format, developing COM-HPC Mini modules that harness Intel Core technology to deliver powerful performance in compact spaces. Similarly, Congatec's upcoming COM-HPC Mini ecosystem is designed to address the needs of advanced embedded systems, including embedded vision, featuring dual MIPI-CSI interfaces for applications with stringent space and power constraints. This development signifies a broader shift towards versatile, high-performance modules that cater to the evolving demands of modern technology applications.

The introduction of COM-HPC Mini modules now enables vendors to consider upgrading existing systems with minimal adjustments. The potential for drone manufacturers, for instance, to adapt to new performance classes without extensive redesigns underscores the modular approach's cost-effectiveness and efficiency. This adaptability is pivotal, particularly when only minor modifications to carrier boards are necessary to meet new standards. This approach underlines the potential for long-term design security and adaptability within the industry, reflecting a shift towards standardising modular design in high-performance computing environments.

As part of the broader COM-HPC family, this compact module offers an ideal solution for space-constrained applications that still require robust computing power. By combining durability, efficient thermal management, and advanced I/O capabilities, the COM-HPC Mini opens up new possibilities in industries, ranging from industrial automation to mobile robotics. Its adaptability and performance in small spaces highlight the ongoing shift toward more flexible, modular design in embedded systems, ensuring that the COM-HPC ecosystem continues to evolve to meet the growing demands of modern technology. **EFY**

Nidhi Agarwal is Senior Technology Journalist at EFY. She is an Electronics and Communication Engineer with over five years of academic experience. Her expertise lies in working with development boards and IoT cloud. She enjoys writing as it enables her to share her knowledge and insights related to electronics, with like-minded techies.

Chargers for Electric Vehicles - Equipped with Diotec MOSFET DIW018N65

Since the transport sector is the only one that uses 25% of the world's energy, it must be changed to a green energy-based travel system to address the growing pollution and global warming challenges. It is also the primary cause of CO2 emissions. Thus, in this case, an EV (Electric Vehicle) might be the best option to address the issues mentioned above. The phenomenon of battery charging requires consideration to optimize an electric vehicle's energy efficiency. An EV battery pulls a peaky amount of current from the source when it is charged using a traditional AC-DC converter, which deteriorates the power profile and indices overall. As a result, the low power quality problems. A power factor correction (PFC) circuit is added to a power supply circuit to bring its power factor close to 1.0 or reduce harmonics.



The active PFC circuit controls the current by turning the MOSFET on and off, synchronizing the power supply voltage and phase, and bringing the input current waveform closer to a sine wave.

The DIW018N65 from Diotech Semiconductor is an 18 A/650V N-channel MOSFET ready to deliver the best switching characteristic, low RDS (on) plays a crucial role while enhancing the system, to meet these conditions, very low on-state resistance of typically 155 mΩ helps improve the heat dissipation performance of MOSFET, very low gate charge (Qg) 38nC to improve the efficiency and improve the power density of MOSFET.

FOR MORE INFO



Diotech Electronics India Private Limited
Survey No.614, Hissa No.3 to 9, Saswad Village,
Purandar Taluka, Dist: Pune, Maharashtra - 412 301
Tel: +91-(0)-2115-225047/222523
Email: info@diotech.in; http://www.diotech.com/